MILITARY SPECIFICATION

SPHEROIDAL PROPELLANT, M38 FOR USE IN PROPELLING CHARGES M218, M219, AND XM220 FOR CARTRIDGES, 81MM, SMOKE, M819, ILLUMINATING M853 AND PRACTICE, XM879

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality acceptance provisions for manufacture and acceptance of Spheroidal Propellant, M38, for use in propelling charges M218, M219, and XM220 for cartridges, 81MM, Smoke, RP, M819, Illuminating M853 and Practice, XM879.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unles otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-D-98	-	Diphenylamine
MIL-G-155	-	Graphite
MIL-P-156	-	Potassium Nitrate
MIL-P-198	-	Potassium Sulfate
MIL-D-204	-	Dinitrotoluene
MIL-P-218	-	Dibutylphthalate
MIL-N-244	-	Nitrocellulose
MIL-N-246	-	Nitroglycerine
MIL-P-293	-	Calcium Carbonate

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-S-50004 - Sodium Sulfate
MIL-S-50005 - Stannic Dioxide

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellant, Solid, Sampling, Examination and Testing
MIL-STD-652 - Propellants, Solid, for Cannon, Requirements and Packing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9278773 - Container assembly
9300234 - Charge, Propelling, M218
9310340 - Cartridge, 81MM, Practice, XM879 with Fuze, Practice, M751
9327839 - Cartridge, 81MM, Smoke, RP, M819
9352621 - Cartridge, 81MM, Illuminating, M853 with Fuze, M768
9378134 - Charge, Propelling, M219
9381510 - Charge, Propelling, XM220

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solcitation.

CODE OF FEDERAL REGULATIONS

49 CFR - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Ingredients. All ingredients used in the manufacture of the propellant, including reworked propellant, shall comply with the applicable specification as follows:

Constituent Materials	Conforming to Specifications
Diphenylamine	MIL-D-98
Graphite	MIL-G-155, Grade IV
Potassium Nitrate	MIL-P-156
Potassium Sulfate	MIL-P-198
Dinitrotoluene	MIL-D-204
Dibutylphthalate	MIL-P-218
Nitrocellulose	MIL-N-244
Nitroglycerine	MIL-N-246
Calcium Carbonate	MIL-C-293
Sodium Sulfate	MIL-S-50004
Stannic Dioxide (Tin Dioxide)	MIL-S-50005

3.2 Chemical requirements

3.2.1 Chemical composition. The chemical composition shall comply with the requirement given in Table I.

TABLE I Composition

Material	Requirement (% by weight)	Applicable Test Method
Nitrocellulose	remainder	4.5.1
(Nitrogen in Nitrocellulose)	13.0 to 13.2	4.5.1
Diphenylamine	0.75 to 1.50	4.5.1
Graphite (max)	0.4	4.5.1
Potassium Nitrate/ Potassium Sulfate	0.85 ± 0.25	4.5.1

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Total Volatiles % (max)	2.0	4.5.1
*Residual Solvent (max)	1.20	4.5.1
Moisture and volatiles %	1.15 ± 0.25	4.5.1

*Added basis.

3.2.2 Hygroscopicity. The value of the hygroscopicity shall not exceed 2.15 percent (measured in the first five (5) production lots only) when determined as specified in 4.5.1.

3.2.3 Chemical stability. During the heat test at 135°C, the propellant shall not explode in less than 5 hours and shall not change standardized methyl violet test paper in less than 40 minutes when determined as specified in 4.5.1.

3.3 Physical characteristics.

3.3.1 Dimensions. The propellant shall have a nominal grain diameter of 0.023 inches when determined as specified in 4.5.2.

3.3.2 Bulk density. The bulk density of the propellant grains shall be 0.930 gram/cc minimum when determined as specified in 4.5.2.

3.3.3 Loadability. The charge weight required to meet the ballistic performance requirements specified in 3.4 shall be capable of being loaded in the propelling charge containers when determined as specified in 4.5.2.

3.4 Ballistics performance requirements. The propellant, when assembled into a test cartridge as shown on drawing 9327839, 9352621 and 9310340 and fired in the M252 Mortar, shall comply with the requirements defined in Table II, III or IV (as applicable).

TABLE II

Ballistic performance requirements for M819

Characteristics	Temperature		
	-50°F	70°F	145°F
Muzzle Velocity (Mean)	--	925 ft/sec	--
Muzzle Velocity Standard Deviation (Max.)	--	6 ft/sec	--
Average Chamber Pressure (Max)	--	12,100 lbs/in^2	--
Individual Chamber Pressure (Max)	--	12,600 lbs/in^2	13,100 lbs/in^2

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TABLE III

Ballistic performance requirements for M853

	-50°F	70°F	145°F
Muzzle Velocity (Mean)	--	1020 ft/sec	--
Muzzle Velocity Standard Deviation (Max)	--	6 ft/sec	--
Average Chamber Pressure (max)	--	12,100 lb/in^2	--
Individual Chamber Pressure (max)	--	12,600 lb/in^2	13,100 lb/in^2

TABLE IV

Ballistic performance requirement for XM879

	-50°F	70°F	145°F
Muzzle Velocity (mean)	--	1005 ft/sec	--
Muzzle Velocity Standard Deviation (max)	--	6 ft/sec	--
Average Chamber Pressure (max)	--	12,100 lbs/in^2	--
Individual Chamber Pressure (max)	--	12,600 lbs/in^2	13,100 lbs/in^2

* Note: Pounds per square inches = lbs/in^2, feet per second = ft/sec.

3.5 Granulation. The propellant shall comply with the requirements of Table V when tested as specified 4.5.2.4.

TABLE V Granulation

US Standard Sieve Size	Percent
18	95 min through
20	90 min through
50	5 max through
60	3 max through

3.6 Reworked propellant. When reworked propellant or any reworked ingredients are used in manufacturing the propellant the following requirements shall apply. The reworked propellant shall be only from approved sources (see 3.7) and the propellant description sheets shall be clearly marked as containing reworked propellant. The final propellant product containing reworked ingredients shall conform to the chemical analysis specification listed in Table I and those listed as follows:

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Material	Requirement (% by weight)	Applicable Test Method
Nitroglycerine (Max)	2.0	4.5.1
Dinitrotoluene (Max)	1.0	4.5.1
Dibutylphthalate (Max)	0.5	4.5.1
Sodium Sulfate (Max)	0.50	4.5.1
Calcium Carbonate (Max)	0.25	4.5.1
Tin Dioxide (Max) (Stannic Dioxide)	0.1	4.5.1
Dust and Foreign matter (Max)	0.3	4.5.1

3.7 Approved sources. The manufacturer shall supply a list of intended sources of reworked propellant to be used in the manufacture of propellant for this specification to the Contracting Officer. This information is to be readily available by the manufacturer to the Government upon request.

3.8 Surveillance. A surveillance test shall be done for each lot of propellant. This information shall be available by the manufacturer to the Government upon request. The surveillance test shall be determined as specified in 4.5.1.

3.9 Workmanship. The best manufacturing practices shall be used in formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard constituent materials shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. Preproduction Sample Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 Preproduction sample inspection. Not applicable.

4.4 Quality conformance inspection. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1 Examination and testing.

4.4.1.1 Sampling for chemical and physical testing. Representative samples shall be taken by random selection of five containers from each propellant lot. Equivalent quantities of the five containers shall be taken such that the aggregate quantity shall be sufficient for the required tests. These samples shall be poured into clean containers, immediately closed with a hermetic seal and identified with a label showing the container number from which the sample was taken, the lot number, propellant nomenclature, and packing date. Prior to conducting lot acceptance tests, these samples shall be permitted to attain room temperature ($70 \pm 5^{o}F$); then equal portions sufficient to make the required quantity of a composite sample shall be taken from each representative sample and blended. The composite sample shall be placed in containers, immediately closed with a hermetic seal, and identified respectively as "Composite Sample". The "Composite Sample" shall be used for chemical and physical tests. If the sample fails to meet any of the requirements specified in section 3, the lot shall be rejected.

4.4.1.2 Chemical and physical characteristics.

4.4.1.2.1 Finished propellant. Each lot of propellant shall be tested for verification of composition and physical properties in accordance with 4.5.1 and 4.5.2.

4.4.1.3 Ballistic performance. Five containers shall be selected at random from each lot of propellant. Equivalent quantities of the five containers shall be taken such that the aggregate quantity shall be sufficient for ballistic performance tests described in 4.5.3.

4.4.1.3.1 Muzzle velocity standard deviation. If the velocity standard deviation of the rounds tested in accordance with 4.5.3.2 exceeds the applicable requirement, apply with retest criteria in accordance with 4.5.3.3. If the retest criteria cited does not apply, the lot shall be rejected.

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4.4.1.3.2 Chamber pressure (average). If the average pressure obtained exceeds the applicable requirement, apply retest criteria in accordance with 4.5.3.3. If the retest criteria cited does not apply, the lot shall be rejected.

4.4.1.3.3 Individual chamber pressure (max). If any individual pressure observed during the uniformity series, or the average chamber pressure plus three times the chamber pressure standard deviation exceeds the applicable requirement, the lot shall be rejected.

4.5 Methods of inspection.

4.5.1 Chemical composition and characteristics (see 6.4). Samples of the propellant shall be subject to the following test methods to verify compliance with applicable requirements.

4.5.1.1 Nitrocellulose. Nitrocellulose content shall be determined in accordance with MIL-STD-286, Method 209.2.1.

4.5.1.2 Dinitrotoluene. Dinitrotoluene content shall be determined in accordance with MIL-STD-286, Method 208.5.1.

4.5.1.3 Diphenylamine. Diphenylamine content shall be determined in accordance with MIL-STD-286, Method 208.5.1.

4.5.1.4 Dibutylphthalate. Dibutylphthalate content shall be determined in accordance with MIL-STD-286, Method 208.5.1.

4.5.1.5 Graphite. Graphite content shall be determined in accordance with MIL-STD-286, Method 306.2.3.

4.5.1.6 Nitroglycerine. Nitroglycerine content shall be determined in accordance with MIL-STD-286, Method 208.5.1

4.5.1.7 Salt. Potassium Nitrate or Potassium Sulfate content shall be determined in accordance with MIL-STD-286, Method 310.4.2.

4.5.1.8 Total volatiles. Total volatile content shall be determined in accordance with MIL-STD-286, Method T103.6 and T 101.6.

4.5.1.9 Residual solvents. Residual solvent shall be determined in accordance with MIL-STD-286, Method 103.4.1.

4.5.1.10 Sodium Sulfate. Sodium sulfate content shall be determined in accordance with MIL-STD-286, Method T316.1.

4.5.1.11 Calcium carbonate. Calcium carbonate content shall be determined in accordance with MIL-STD-286, Method T316.1.

4.5.1.12 Nitrogen. Nitrogen in the nitrocellulose shall be determined in accordance with MIL-STD-286, Method T209.9.

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4.5.1.13 Hygroscopicity. Hygroscopicity shall be determined in accordance with MIL-STD-286, Method 503.1.3.

4.5.1.14 Stability. Chemical stability of the propellant shall be determined in accordance with MIL-STD-286, Method 404.1.2.

4.5.1.15 Moisture and volatiles. Moisture and volatiles shall be determined in accordance with MIL-STD-286, Method 101.2.2.

4.5.1.16 Tin Dioxide (Stannic Dioxide). Tin dioxide shall be determined in accordance with MIL-STD-286 Method 306.2.3.

4.5.1.17 Dust and foreign matter. Dust and foreign matter content shall be determined in accordance with MIL-STD-286 Method 501.1.3.

4.5.1.18 Surveillance test. The surveillance test shall be determined in accordance with MIL-STD-286, Method 407.1

4.5.2 Physical characteristics (see 6.4).

4.5.2.1 Dimensions. Grain diameter shall be calculated from the method 506.1, MIL-STD-286.

4.5.2.2 Bulk density. The bulk density shall be determined in accordance with MIL-STD-286, Method 502.1.1.

4.5.2.3 Loadability. Loadability characteristics shall be verified by meeting the requirements of 4.5.3.

4.5.2.4 Granulation. The granulation shall be determined in accordance with MIL-STD-286, Method 506.1.

4.5.3 Ballistic performance. These ballistic tests shall be conducted at a proving ground in accordance with the applicable test procedures. The cartridges shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning. Assurance shall be made that, when fired, the rounds are at the conditioning temperature for a minimum of 24 hours. The following tolerances shall apply to the following temperature conditioning:

$-50 + 5^{\circ}F$
$70^{\circ} \pm 2.5^{\circ}F$
$145 - 5^{\circ}F$

The cartridges shall be loaded and inspected by LAP facility in accordance with drawing 9327839 or 9352621 or 9310340 (or appropriate designated alternate as specified by the contracting officer), and shipped to an approved proving ground for ballistic testing. The weight of the projectile shall be as follows:

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CARTRIDGE	PROP CHARGE	PROJECTILE WEIGHT
M819	M218	10.35 ± 0.02 lbs
M853	M219	8.90 ± 0.02 lbs
XM879	XM220	9.00 ± 0.02 lbs

4.5.3.1 Charge establishment. Three groups of three cartridges each loaded with selected charge weight of propellant (weights selected should bracket the expected service charge weight) shall be conditioned and fired at 70°F in the M252 Mortar. Three control cartridges shall also be conditioned and fired at 70°F in the M252 Mortar. From the data obtained, the charge weight vs pressure vs velocity curve shall be established. A charge weight shall be calculated which will yield the prescribed service velocity (see 3.4). This established charge weight shall then be loaded into cartridges for the uniformity series (see 4.5.3.2).

4.5.3.2 Uniformity. Thirty cartridges shall be assembled with the established charge weight of propellant. Three groups of ten cartridges each shall be conditioned to -50°F, 70°F, and 145°F, respectively, and fired in the M252 Mortar. Cartridges fired at 70°F shall be fired alternately with control cartridge rounds. The ten control cartridges shall be conditioned at 70°F. The mean velocity and mean pressure and standard deviation of velocity shall be calculated. The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated by a final adjustment the charge weight using the slope of the velocity charge weight curve established in 4.5.3.1. The resultant will be the recommended charge weight.

4.5.3.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary:

a. If, for any reason, the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

b. If the recommended charge weight exceeds 2% of the charge weight used in the uniformity series, repeat the charge establishment and the uniformity series at 70°F.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. There is no available approved Level A Pack.

5.1.2 Level B. There is no available approved Level B Pack.

5.1.3 Level C. Pack 60 Pounds net weight maximum in a DOT-21-C-115 Fiber Drum (h=21 3/4", d=14") in accordance with Title 49 CFR Section 173.93.

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5.2 Marking.

5.2.1 Level A. Not available

5.2.2 Level B. Not available

5.2.3 Level C. Marking shall be in accordance with Title 49 CFR Section 173.93 and MIL-STD-129. DOT Nomenclature shall be 'Propellant Explosives (Solid) Class B'.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use in Cartridge, 81MM, Smoke, RP, M819, illuminating M853, and practice XM879.

6.2 Ordering data. See MIL-A-48078.

6.3 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description Sheet shall be forwarded to Commander, US Army Armament Research, Development, and Engineering Center (ARDEC), ATTN: AMSMC-QAR-R (D), and one copy at ATTN: SMCAR-AEE-BP, Dover, NJ 07801-5001.

6.4 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: ARDEC, ATTN: AMSMC-QAR-R (D), Dover, NJ 07801-5001. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5 Surveillance sample. A five pound sample of propellant shall be forwarded to US Army Armament Research, Development and Engineering Center (ARDEC), Dover, NJ, 07801-5001, ATTN: SMCAR-LCE-P for the propellant surveillance test program in accordance with MIL-STD-652.

6.6 Subject term (key word) listing.

Ammunition
Mortar ammunition
Powder, ball
Propellant

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1376-A263)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-70547	SPHEROIDAL PROPELLANT, M38

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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